MATH 8090:
State-Space Models I

Hidden States, Noisy Observations, and the Kalman Filter

Readings: SS17 Chapters 6.1–6.2; BD Chapters 9.1–9.3

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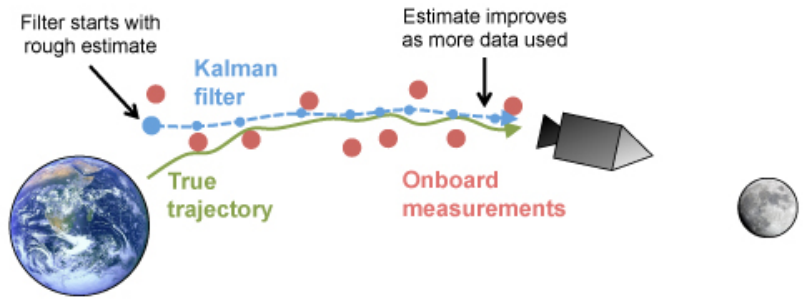


School of
MATHEMATICAL AND
STATISTICAL SCIENCES
Clemson University

Week 13

Historical Motivation: Spacecraft Tracking

State-space models originated in tracking and control [Kalman, 1960; Kalman and Bucy, 1961].



- The spacecraft's **state** includes its position and velocity.
- Its motion follows a dynamical model, subject to uncertainty.
- Tracking instruments provide **noisy observations** of the system.

How can we infer the evolving hidden state from imperfect observations?

Two Sources of Information

To estimate the spacecraft's position, we can use:

1. **Knowledge of its motion**
   Physical laws help predict where the spacecraft should move next.

2. **New measurements**
   Tracking instruments provide information about its current position.

Prediction from the model $+$ **Information from measurements**

$\Downarrow$

An improved estimate of the current state

Historical Origins: Kalman's Insight

- In 1960, Rudolf Kalman introduced a recursive approach to filtering and prediction.
- Kalman and Bucy (1961) extended the framework to continuous-time systems.
- The central idea is to combine two sources of information:
  1. **System dynamics:** How the hidden state evolves.
  2. **Observations:** What noisy measurements tell us.

Predict the state using the model,
then correct the prediction using new data.

General State-Space Framework

A state-space model describes a system through two components:

1. **State dynamics:** How does the hidden state evolve?

$$\boxed{\boldsymbol{X}_t = f_t(\boldsymbol{X}_{t-1}, \boldsymbol{V}_t)}$$

2. **Observation process:** How is the state measured?

$$\boxed{\boldsymbol{Y}_t = h_t(\boldsymbol{X}_t, \boldsymbol{W}_t)}$$

- $\boldsymbol{X}_t$: latent state
- $\boldsymbol{Y}_t$: observed data
- $\boldsymbol{V}_t$, $\boldsymbol{W}_t$: process and measurement noise

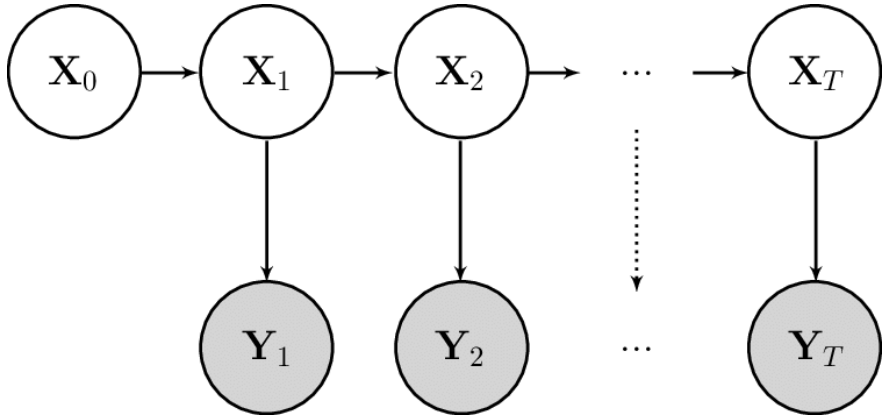
The state evolves over time but is observed only indirectly.

From State-Space Models to Kalman Filtering

1. **Modeling:** Describe hidden states and noisy observations.
2. **Prediction:** Use system dynamics to predict the next state.
3. **Updating:** Combine the prediction with new measurements.
4. **Inference:** Estimate hidden states and quantify uncertainty.

Predict $\longrightarrow$ **Observe** $\longrightarrow$ **Update** $\longrightarrow$ **Repeat**

Linear State-Space Model



$$\text{State:} \quad \boldsymbol{X}_t = \boldsymbol{M}_t \boldsymbol{X}_{t-1} + \boldsymbol{V}_t, \qquad \boldsymbol{V}_t \sim \mathrm{WN}(\boldsymbol{0}, \boldsymbol{Q}_t),$$

$$\text{Observation:} \quad \boldsymbol{Y}_t = \boldsymbol{H}_t \boldsymbol{X}_t + \boldsymbol{W}_t, \qquad \boldsymbol{W}_t \sim \mathrm{WN}(\boldsymbol{0}, \boldsymbol{R}_t).$$

- $\boldsymbol{X}_t \in \mathbb{R}^p$: latent state; $\boldsymbol{Y}_t \in \mathbb{R}^q$: observations.
- $\boldsymbol{M}_t \in \mathbb{R}^{p \times p}$: state transition; $\boldsymbol{H}_t \in \mathbb{R}^{q \times p}$: observation mapping.
- $\boldsymbol{V}_t$ and $\boldsymbol{W}_t$: state and observation noises, respectively.

Assumptions and Dependence Structure

Noise processes

$$E(\boldsymbol{V}_t) = \mathbf{0}, \quad \mathrm{Cov}(\boldsymbol{V}_t) = \boldsymbol{Q}_t,$$
$$E(\boldsymbol{W}_t) = \mathbf{0}, \quad \mathrm{Cov}(\boldsymbol{W}_t) = \boldsymbol{R}_t.$$

1. **Temporal uncorrelatedness**

$$\mathrm{Cov}(\boldsymbol{V}_t, \boldsymbol{V}_s) = \mathrm{Cov}(\boldsymbol{W}_t, \boldsymbol{W}_s) = \mathbf{0}, \qquad t \neq s.$$

2. **State and observation noise**

$$\mathrm{Cov}(\boldsymbol{V}_t, \boldsymbol{W}_s) = \mathbf{0}, \qquad \forall\, t, s.$$

3. **Initial state**

$$E(\boldsymbol{X}_0) = \boldsymbol{\mu}_0, \qquad \mathrm{Cov}(\boldsymbol{X}_0) = \boldsymbol{\Sigma}_0,$$

$$\mathrm{Cov}(\boldsymbol{X}_0, \boldsymbol{V}_t) = \mathrm{Cov}(\boldsymbol{X}_0, \boldsymbol{W}_t) = \mathbf{0}.$$

Forecasting, Filtering, and Smoothing

Goal: Infer the hidden state $\boldsymbol{X}_t$ using observations available through time s:

$$\widehat{\boldsymbol{X}}_{t|s} = E(\boldsymbol{X}_t \mid \boldsymbol{Y}_{1:s}), \qquad \boldsymbol{Y}_{1:s} = \{\boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_s\}.$$

1. **Forecasting** $(s < t)$
   Estimate a future state using past observations.

2. **Filtering** $(s = t)$
   Estimate the current state using observations up to now.

3. **Smoothing** $(s > t)$
   Re-estimate a past state using later observations.

Same state-space model; different information sets.

Applications and Extensions

Many familiar time-series models admit state-space representations:

- ARIMA and seasonal ARIMA: autoregressive dynamics and latent components.
- Vector autoregression (VAR): multivariate state dynamics.
- Structural time-series models: trend, seasonal, and other latent components.
- Hidden Markov models (HMMs): discrete latent states with probabilistic transitions.

Computational advantages

- Recursive state estimation and forecasting.
- Natural treatment of missing observations.

From AR(1) to ARMA(1,1)

Recall the stationary AR(1) model:

$$X_t = \phi X_{t-1} + Z_t, \qquad Z_t \sim \mathrm{WN}(0, \sigma^2), \quad |\phi| < 1.$$

Its state-space representation is immediate:

$$\underbrace{X_t = \phi X_{t-1} + Z_t}_{\text{State equation}}, \qquad \underbrace{Y_t = X_t}_{\text{Observation equation}}$$

ARMA(1,1):

$$Y_t - \phi Y_{t-1} = Z_t + \theta Z_{t-1}, \qquad |\phi| < 1.$$

Define $X_t = (1 - \phi B)^{-1} Z_t$. Then

$$\boxed{Y_t = (1 + \theta B) X_t = X_t + \theta X_{t-1}.}$$

ARMA(1,1): State-Space Representation

Augmented state:

$$\boldsymbol{X}_t = \begin{bmatrix} X_t \\ X_{t-1} \end{bmatrix}$$

State equation:

$$\boldsymbol{X}_t = \underbrace{\begin{bmatrix} \phi & 0 \\ 1 & 0 \end{bmatrix}}_{\boldsymbol{M}} \boldsymbol{X}_{t-1} + \underbrace{\begin{bmatrix} Z_t \\ 0 \end{bmatrix}}_{\boldsymbol{V}_t}$$

Observation equation:

$$Y_t = \underbrace{\begin{bmatrix} 1 & \theta \end{bmatrix}}_{\boldsymbol{H}} \boldsymbol{X}_t, \qquad W_t = 0$$

Noise covariance:

$$\boldsymbol{Q} = \sigma^2 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad R = 0$$

ARMA(1,1): Stationary Initialization

Stationary AR(1) state:

$$X_t = \sum_{j=0}^{\infty} \phi^j Z_{t-j}, \qquad |\phi| < 1.$$

Initial state and covariance:

$$\boldsymbol{X}_0 = \begin{bmatrix} X_0 \\ X_{-1} \end{bmatrix}, \qquad E(\boldsymbol{X}_0) = \mathbf{0},$$

$$\boxed{\boldsymbol{\Sigma}_0 = \frac{\sigma^2}{1-\phi^2} \begin{bmatrix} 1 & \phi \\ \phi & 1 \end{bmatrix}}$$

Stationarity condition:

$$\boldsymbol{\Sigma}_0 = \boldsymbol{M}\boldsymbol{\Sigma}_0\boldsymbol{M}^T + \boldsymbol{Q}.$$

$\boldsymbol{X}_0$ is uncorrelated with future innovations Z_t, $t \geq 1$.

Local-Level Model

A latent, time-varying level observed with noise:

State equation:

$$X_t = X_{t-1} + V_t, \qquad V_t \overset{iid}{\sim} N(0, \sigma_V^2).$$

Observation equation:

$$Y_t = X_t + W_t, \qquad W_t \overset{iid}{\sim} N(0, \sigma_W^2).$$

Initial state:

$$X_0 \sim N(\mu_0, \sigma_0^2).$$

Assume X_0, $\{V_t\}$, and $\{W_t\}$ are mutually independent.

- X_t: random walk (evolving level).
- Y_t: noisy measurement of the level.

Local-Level Model: Signal-to-Noise Ratio

Random-walk dynamics:

$$X_t = X_{t-1} + V_t, \qquad E(X_t \mid X_{t-1}) = X_{t-1}.$$

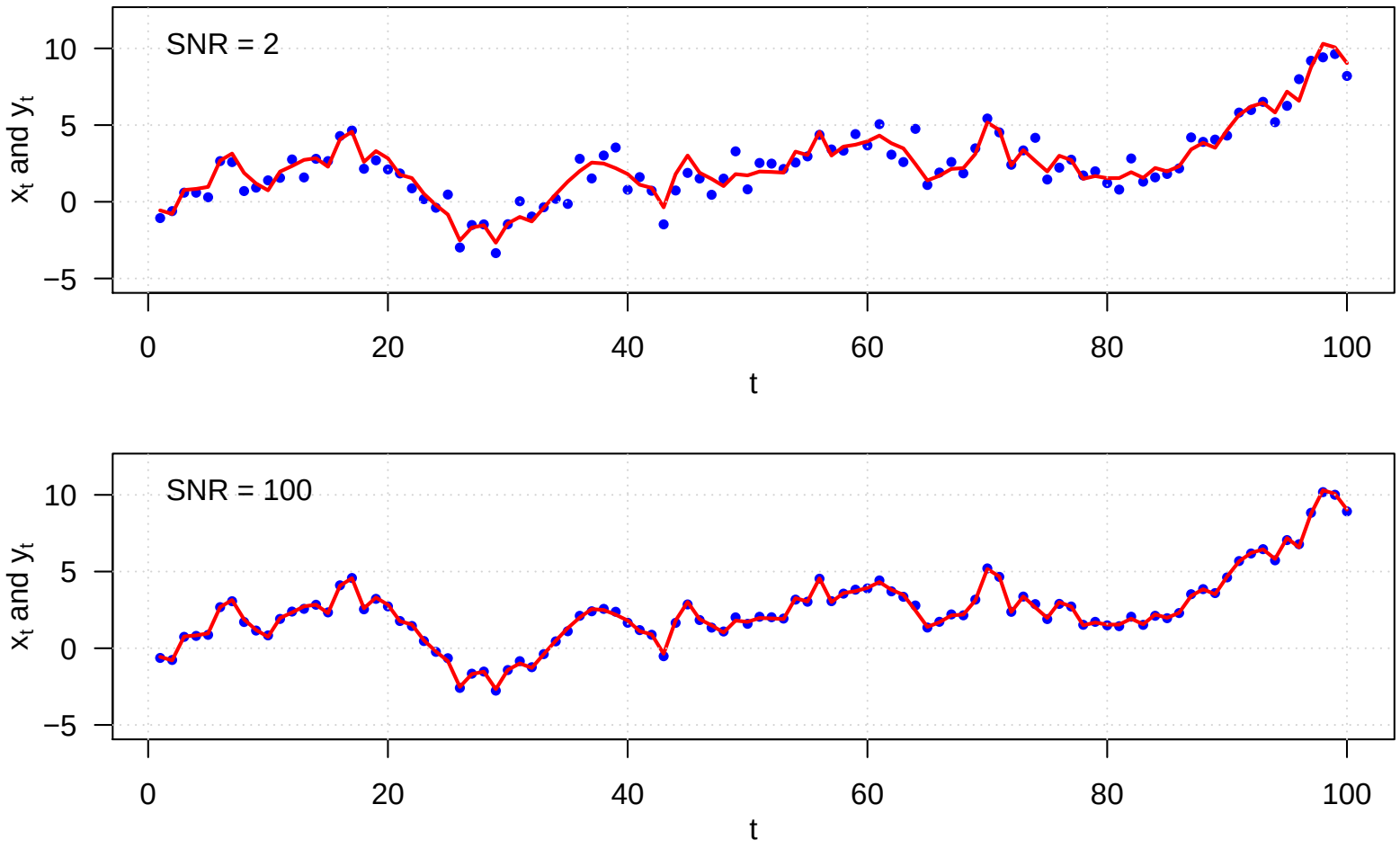
Signal-to-noise ratio:

$$\boxed{\text{SNR} = \frac{\sigma_V^2}{\sigma_W^2}}$$

- High SNR: the latent level changes substantially relative to measurement noise.
- Low SNR: the latent level changes slowly relative to measurement noise.

Question: How much should we trust a new observation relative to the predicted state?

Local-Level Model: Different SNRs



Local-Level Model: Forecasting and Filtering

Goal: Estimate X_t using available observations.

Forecasting:

$$X_t \mid \boldsymbol{y}_{1:t-1} \sim N(\mu_t^f, \Sigma_t^f).$$

Filtering:

$$X_t \mid \boldsymbol{y}_{1:t} \sim N(\mu_t^a, \Sigma_t^a).$$

- μ_t^f, Σ_t^f: forecast mean and MSE.
- μ_t^a, Σ_t^a: filtered mean and MSE.

Forecast $\longrightarrow$ Observe $\longrightarrow$ Update

Kalman Filter: Local-Level Model

Suppose

$$X_{t-1} \mid y_{1:t-1} \sim N(\mu^a_{t-1}, \Sigma^a_{t-1}).$$

1. Prediction

$$\mu^f_t = \mu^a_{t-1}, \qquad \Sigma^f_t = \Sigma^a_{t-1} + \sigma^2_V.$$

2. Kalman gain

$$\boxed{K_t = \frac{\Sigma^f_t}{\Sigma^f_t + \sigma^2_W}}$$

3. Update

$$\mu^a_t = \mu^f_t + K_t(y_t - \mu^f_t),$$

$$\Sigma^a_t = (1 - K_t)\Sigma^f_t.$$

Interpretation: Larger measurement uncertainty reduces the weight assigned to the new observation.

Linear Gaussian State-Space Model

State equation:

$$\boldsymbol{X}_t = \boldsymbol{M}_t \boldsymbol{X}_{t-1} + \boldsymbol{V}_t, \qquad \boldsymbol{V}_t \sim N(\boldsymbol{0}, \boldsymbol{Q}_t).$$

Observation equation:

$$\boldsymbol{Y}_t = \boldsymbol{H}_t \boldsymbol{X}_t + \boldsymbol{W}_t, \qquad \boldsymbol{W}_t \sim N(\boldsymbol{0}, \boldsymbol{R}_t).$$

Initial distribution:

$$\boldsymbol{X}_0 \sim N(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0).$$

Assume $\boldsymbol{X}_0$, $\{\boldsymbol{V}_t\}$, and $\{\boldsymbol{W}_t\}$ are mutually independent, with independent noise sequences over time.

Key consequence: Prediction and filtering distributions remain Gaussian.

Kalman Filter: General Recursions

Suppose

$$\boldsymbol{X}_{t-1} \mid \boldsymbol{y}_{1:t-1} \sim N(\boldsymbol{\mu}_{t-1}^a, \boldsymbol{\Sigma}_{t-1}^a).$$

1. Prediction

$$\boldsymbol{\mu}_t^f = \boldsymbol{M}_t \boldsymbol{\mu}_{t-1}^a,$$

$$\boldsymbol{\Sigma}_t^f = \boldsymbol{M}_t \boldsymbol{\Sigma}_{t-1}^a \boldsymbol{M}_t^T + \boldsymbol{Q}_t.$$

2. Kalman gain

$$\boldsymbol{K}_t = \boldsymbol{\Sigma}_t^f \boldsymbol{H}_t^T (\boldsymbol{H}_t \boldsymbol{\Sigma}_t^f \boldsymbol{H}_t^T + \boldsymbol{R}_t)^{-1}.$$

3. Update

$$\boldsymbol{\mu}_t^a = \boldsymbol{\mu}_t^f + \boldsymbol{K}_t(\boldsymbol{y}_t - \boldsymbol{H}_t \boldsymbol{\mu}_t^f),$$

$$\boldsymbol{\Sigma}_t^a = (\boldsymbol{I} - \boldsymbol{K}_t \boldsymbol{H}_t) \boldsymbol{\Sigma}_t^f.$$

Four Inference Problems in State-Space Models

Consider the local-level model

$$X_t = X_{t-1} + V_t, \qquad Y_t = X_t + W_t,$$

with observations $Y_{1:t} = (Y_1, \ldots, Y_t)$.

Our main questions:

1. Filtering: Estimate the current state X_t using $Y_{1:t}$.
2. Forecasting: Predict the future state X_{t+1} using $Y_{1:t}$.
3. Smoothing: Estimate a past state X_s, $s < t$, using $Y_{1:t}$.
4. Parameter estimation: Estimate σ_V^2, σ_W^2 and the initial-state parameters.

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Our focus: Filtering and forecasting under minimum mean squared error (MSE).

From Prediction to Updating

Suppose we have observed $Y_{1:t-1}$ and obtained a prediction of the hidden state X_t.

Two sources of information:

- Prior prediction: What do past observations tell us about X_t?
- New measurement: What additional information does Y_t provide?

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The key question: How should we combine the prior prediction with the new noisy observation?

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The key question: How should we combine the prior prediction with the new noisy observation?

Assume the initial state and innovations are independent Gaussian random variables.

Joint Gaussian distribution $\Longrightarrow$ Conditional Gaussian distribution

This leads directly to the Kalman filtering update.

Gaussian Regression Lemma

Suppose $\boldsymbol{X}$ and $\boldsymbol{Y}$ are jointly Gaussian:

$$\begin{bmatrix} \boldsymbol{X} \\ \boldsymbol{Y} \end{bmatrix} \sim N\left(\begin{bmatrix} \boldsymbol{\mu}_X \\ \boldsymbol{\mu}_Y \end{bmatrix}, \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{bmatrix} \right),$$

where Σ_{YY} is nonsingular.

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where Σ_{YY} is nonsingular.

Conditional mean (best MSE predictor):

$$\boxed{E(\boldsymbol{X} \mid \boldsymbol{Y}) = \boldsymbol{\mu}_X + \Sigma_{XY}\Sigma_{YY}^{-1}(\boldsymbol{Y} - \boldsymbol{\mu}_Y)}$$

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Conditional mean (best MSE predictor):

$$\boxed{E(\boldsymbol{X} \mid \boldsymbol{Y}) = \boldsymbol{\mu}_X + \Sigma_{XY}\Sigma_{YY}^{-1}(\boldsymbol{Y} - \boldsymbol{\mu}_Y)}$$

Conditional covariance (prediction uncertainty):

$$\boxed{\mathrm{Var}(\boldsymbol{X} \mid \boldsymbol{Y}) = \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}}$$

Interpretation: Prediction = prior mean + correction from new information.

From Gaussian Conditioning to Kalman Filtering

Consider the local-level model:

$$Y_t = X_t + W_t, \qquad W_t \sim N(0, \sigma_W^2).$$

Given past observations $\boldsymbol{Y}_{1:t-1}$, suppose

$$X_t \mid \boldsymbol{Y}_{1:t-1} \sim N(\mu_t^f, \Sigma_t^f).$$

The conditional moments are

$$\begin{aligned} E(X_t \mid \boldsymbol{Y}_{1:t-1}) &= \mu_t^f, \\ E(Y_t \mid \boldsymbol{Y}_{1:t-1}) &= \mu_t^f, \\ \mathrm{Cov}(X_t, Y_t \mid \boldsymbol{Y}_{1:t-1}) &= \Sigma_t^f, \\ \mathrm{Var}(Y_t \mid \boldsymbol{Y}_{1:t-1}) &= \Sigma_t^f + \sigma_W^2. \end{aligned}$$

Applying the Gaussian regression lemma:

$$\boxed{\mu_t^a = \mu_t^f + \underbrace{\frac{\Sigma_t^f}{\Sigma_t^f + \sigma_W^2}}_{K_t}(Y_t - \mu_t^f)}$$

Regression Corollary: Scalar Response

Suppose $(X, \boldsymbol{Y}^T)^T$ is jointly Gaussian, where X is scalar.

Conditional distribution:

$$X \mid \boldsymbol{Y} = \boldsymbol{y} \sim N\left(\mu_{X|\boldsymbol{y}}, \sigma^2_{X|\boldsymbol{Y}}\right),$$

where

$$\boxed{\begin{aligned} \mu_{X|\boldsymbol{y}} &= \mu_X + \boldsymbol{\Sigma}_{X\boldsymbol{Y}}\boldsymbol{\Sigma}^{-1}_{\boldsymbol{Y}\boldsymbol{Y}}(\boldsymbol{y} - \boldsymbol{\mu}_{\boldsymbol{Y}}), \\ \sigma^2_{X|\boldsymbol{Y}} &= \sigma^2_X - \boldsymbol{\Sigma}_{X\boldsymbol{Y}}\boldsymbol{\Sigma}^{-1}_{\boldsymbol{Y}\boldsymbol{Y}}\boldsymbol{\Sigma}_{\boldsymbol{Y}X}. \end{aligned}}$$

- The conditional mean is the optimal mean-square predictor.
- The conditional variance is its mean squared prediction error (MSE).

Connection to Time-Series Prediction

Let $\{X_t\}$ be a zero-mean stationary Gaussian process with autocovariance $\gamma(h)$.

To predict X_{n+1}, define

$$\boldsymbol{Y} = (X_n, \ldots, X_1)^T, \qquad \boldsymbol{\gamma}_n = \begin{bmatrix} \gamma(1) \\ \vdots \\ \gamma(n) \end{bmatrix},$$

where

$$\boldsymbol{\Gamma}_n = \mathrm{Var}(\boldsymbol{Y}), \qquad [\boldsymbol{\Gamma}_n]_{ij} = \gamma(i-j).$$

Best linear predictor:

$$\boxed{\widehat{X}_{n+1} = \boldsymbol{\gamma}_n^T \boldsymbol{\Gamma}_n^{-1} \boldsymbol{Y}}$$

Prediction MSE:

$$\boxed{v_n = \gamma(0) - \boldsymbol{\gamma}_n^T \boldsymbol{\Gamma}_n^{-1} \boldsymbol{\gamma}_n}$$

Time-series prediction is a special case of Gaussian regression.

Filtering for Local-Level Model: I

Recall the local-level model:

$$Y_t = X_t + W_t, \qquad W_t \overset{iid}{\sim} N(0, \sigma_W^2),$$
$$X_t = X_{t-1} + V_t, \qquad V_t \overset{iid}{\sim} N(0, \sigma_V^2).$$

Assume

$$X_0 \sim N(\mu_0, \sigma_0^2),$$

independent of the two mutually independent noise sequences.

Filtering problem: Estimate the hidden state X_t using observations

$$\boldsymbol{Y}_{1:t} = (Y_1, \ldots, Y_t)^T.$$

Filtering for Local Level Model: II

The filtered estimate of X_t given $\boldsymbol{Y}_{1:t}$ is

$$\boxed{\mu_t^a = E(X_t \mid \boldsymbol{Y}_{1:t}) = \mu_t + \boldsymbol{\Sigma}_{t,t}^T \boldsymbol{\Sigma}_{Y,t}^{-1} (\boldsymbol{Y}_{1:t} - \boldsymbol{\mu}_{1:t})}$$

where

- $\mu_t = \mathbb{E}(X_t)$, $\boldsymbol{\mu}_{1:t}$ is a vector containing, for $j = 1, \cdots, t$,

$$\mu_j = \mathbb{E}(X_j) = \mathbb{E}(X_j + W_j) = \mathbb{E}(Y_j)$$

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- Vector $\Sigma_{t,t}$ contains covariances between X_t and $\mathbf{Y}_{1:t}$

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- Vector $\Sigma_{t,t}$ contains covariances between X_t and $\mathbf{Y}_{1:t}$
- (i, j)th element of matrix $\Sigma_{Y,t}$ is covariance between Y_i and Y_j

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- Note: $\mathbb{E}(\mu_t^a) = \mathbb{E}[\mathbb{E}(X_t | \mathbf{Y}_{1:t})] = \mathbb{E}(X_t) = \mu_t$

Filtering for Local Level Model: II

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- Note: $\mathbb{E}(\mu_t^a) = \mathbb{E}[\mathbb{E}(X_t|\mathbf{Y}_{1:t})] = \mathbb{E}(X_t) = \mu_t$
- With $\sigma_t^2 = \mathrm{Var}(X_t)$, MSE for predictor is

$$\boxed{\mathbb{E}[(X_t - \mu_t^a)^2] = \sigma_t^2 - \Sigma_{t,t}^T \Sigma_{Y,t}^{-1} \Sigma_{t,t} = \Sigma_t^a}$$

Forecasting for Local-Level Model: I

Forecasting: Estimate X_{t+1} given $\boldsymbol{Y}_{1:t}$.

- The optimal forecast is

$$\boxed{\mu_{t+1}^{f} = \mu_{t+1} + \boldsymbol{\Sigma}_{t+1,t}^{T} \boldsymbol{\Sigma}_{Y,t}^{-1} (\boldsymbol{Y}_{1:t} - \boldsymbol{\mu}_{1:t})}$$

where $\boldsymbol{\Sigma}_{t+1,t} = \mathrm{Cov}(\boldsymbol{Y}_{1:t}, X_{t+1})$.

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- By iterated expectation,

$$E(\mu_{t+1}^f) = E(X_{t+1}) = \mu_{t+1}.$$

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  where $\boldsymbol{\Sigma}_{t+1,t} = \mathrm{Cov}(\boldsymbol{Y}_{1:t}, X_{t+1})$.

- By iterated expectation,

$$E(\mu_{t+1}^f) = E(X_{t+1}) = \mu_{t+1}.$$

- The forecast MSE is

$$\boxed{\Sigma_{t+1}^f = \sigma_{t+1}^2 - \boldsymbol{\Sigma}_{t+1,t}^T \boldsymbol{\Sigma}_{Y,t}^{-1} \boldsymbol{\Sigma}_{t+1,t}.}$$

Forecasting for Local-Level Model: II

Consider forecasting Y_{t+1} given $\boldsymbol{Y}_{1:t}$:

$$\widehat{Y}_{t+1|t} = E(Y_{t+1} \mid \boldsymbol{Y}_{1:t}).$$

Since $Y_{t+1} = X_{t+1} + W_{t+1}$ and W_{t+1} is independent of past observations,

$$\mathrm{Cov}(Y_{t+1}, Y_j) = \mathrm{Cov}(X_{t+1}, Y_j), \quad j \leq t.$$

Thus,

$$\boxed{\widehat{Y}_{t+1|t} = E(X_{t+1} \mid \boldsymbol{Y}_{1:t}) = \mu^f_{t+1}.}$$

Same forecast, but different MSEs:

$$\boxed{\mathrm{MSE}(\widehat{Y}_{t+1|t}) = \Sigma^f_{t+1} + \sigma^2_W.}$$

Filtering for Local-Level Model: III

To compute μ_t^a and its MSE Σ_t^a, we need:

1. $\mu_j = E(X_j)$, $j = 1, \ldots, t$.
2. $\mathrm{Cov}(X_t, Y_j)$, $j = 1, \ldots, t$.
3. $\mathrm{Cov}(Y_j, Y_k)$, $1 \leq j, k \leq t$.
4. $\sigma_t^2 = \mathrm{Var}(X_t)$.

From the state and observation equations,

$$\boxed{X_j = X_0 + \sum_{l=1}^{j} V_l, \qquad Y_j = X_0 + \sum_{l=1}^{j} V_l + W_j.}$$

Filtering for Local-Level Model: IV

Using

$$X_j = X_0 + \sum_{l=1}^{j} V_l, \qquad Y_j = X_0 + \sum_{l=1}^{j} V_l + W_j,$$

we obtain

$$\mu_j = E(X_j) = E(Y_j) = \mu_0.$$

For $j \leq t$,

$$\begin{aligned} \mathrm{Cov}(X_t, Y_j) &= \mathrm{Cov}\left(X_0 + \sum_{l=1}^{t} V_l,\ X_0 + \sum_{l=1}^{j} V_l + W_j \right) \\ &= \boxed{\sigma_0^2 + j\sigma_V^2}. \end{aligned}$$

For $j, k \leq t$,

$$\boxed{\mathrm{Cov}(Y_j, Y_k) = \sigma_0^2 + \min(j,k)\sigma_V^2 + \delta_{jk}\sigma_W^2.}$$

$\delta_{jk} = 1$ if $j = k$, and 0 otherwise.

Filtering for Local-Level Model: V

Since

$$X_t = X_0 + \sum_{l=1}^{t} V_l,$$

we have

$$\boxed{\sigma_t^2 = \mathrm{Var}(X_t) = \sigma_0^2 + t\sigma_V^2.}$$

- We now have all the quantities needed to compute the filtered estimate μ_t^a and its MSE Σ_t^a.
- Similar calculations yield the forecast μ_{t+1}^f and its MSE Σ_{t+1}^f.

Kalman Recursions for Filtering/Forecasting: I

The regression formulas give

$$\mu_t^a = \mu_t + \boldsymbol{\Sigma}_{t,t}^T \boldsymbol{\Sigma}_{Y,t}^{-1}(\boldsymbol{Y}_{1:t} - \boldsymbol{\mu}_{1:t}),$$

and

$$\mu_{t+1}^f = \mu_{t+1} + \boldsymbol{\Sigma}_{t+1,t}^T \boldsymbol{\Sigma}_{Y,t}^{-1}(\boldsymbol{Y}_{1:t} - \boldsymbol{\mu}_{1:t}).$$

Computational challenge: $\boldsymbol{\Sigma}_{Y,t}$ is a $t \times t$ matrix, whose dimension increases with t.

The Kalman recursions avoid repeated inversion of the full observation covariance matrix.

Key idea: Update the state estimate and its MSE recursively as new observations arrive.

Kalman Recursions for Filtering/Forecasting: II

At time t, the new observation Y_t becomes available.

The Kalman recursion takes

$$\boxed{\mu_t^f, \quad \Sigma_t^f, \quad Y_t}$$

as inputs and produces

$$\boxed{\mu_t^a, \quad \Sigma_t^a, \quad \mu_{t+1}^f, \quad \Sigma_{t+1}^f.}$$

Six recursive steps:

1. Steps 1–2: Innovation and its MSE.
2. Steps 3–4: Filtering.
3. Steps 5–6: Forecasting.

Kalman Recursions for Filtering/Forecasting: III

1. **Innovation**

$$U_t = Y_t - \widehat{Y}_{t|t-1} = Y_t - \mu_t^f.$$

2. **Innovation variance**

$$F_t = \Sigma_t^f + \sigma_W^2.$$

3. **Filtered estimate**

$$\boxed{\mu_t^a = \mu_t^f + K_t U_t, \qquad K_t = \frac{\Sigma_t^f}{F_t}.}$$

where K_t is the Kalman gain.

4. **Filtering MSE**

$$\boxed{\Sigma_t^a = (1 - K_t)\Sigma_t^f.}$$

Kalman Recursions for Filtering/Forecasting: IV

5. One-step state forecast

$$\boxed{\mu_{t+1}^f = \mu_t^a = \mu_t^f + K_t U_t.}$$

6. Forecast MSE

$$\boxed{\Sigma_{t+1}^f = \Sigma_t^a + \sigma_V^2 = (1 - K_t)\Sigma_t^f + \sigma_V^2.}$$

Initialization:

$$\mu_1^f = \mu_0, \qquad \Sigma_1^f = \sigma_0^2 + \sigma_V^2.$$

Apply steps 1–6 sequentially for $t = 1, \ldots, n$.